

Field Symmetries

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Take L/K a finite extension. Goal is to bound $G := \text{Gal}(L/K)$.

Consider another field F . Let $g : K \rightarrow F$ be a map of rings.

Let $\text{Ext}_K^L(g) := \{h : L \rightarrow F \mid h|_K = g\}$.

Problem 1. Show that if $F = L$, and g the original inclusion, then this Ext is just G .

Proof. $\forall h \in \text{Ext}_K^L(g)$, h is a inclusion and thus a isomorphism of rings (more specifically automorphism). If g is the original inclusion, it is a identity map. Thus $h|_K = \text{id}_K$. Thus by definition of Galois Group, $h \in G$. Thus $\text{Ext}_K^L(g) \subseteq G \forall h \in G$, $h|_K = \text{id}_K = g$, by definition of Galois Group. Thus $h \in \text{Ext}_K^L(g)$. Thus $G \subseteq \text{Ext}_K^L(g)$. Thus $\text{Ext}_K^L(g) = G$. $\square$

Problem 2a. Suppose $L = K[a]$, and $p(X) \in K[X]$ is the minimal monic poly. Construct a canonical bijection between Ext and the roots of $g(p)[X]$ in $F[X]$ (note g induces $L[X] \rightarrow F[X]$, a ring homo, just applying g to the coefficients and sending X to X).

Proof. The canonical map between $\text{Ext}_K^L(g)$ and the roots of $g(p)(X)$ is the following $h \rightarrow h(a)$.

Suppose $\forall h, k \in \text{Ext}_K^L(g)$, $h(a) = k(a) \implies \forall x+ya, h(x+ya) = g(x)+g(y)h(a) = g(x)+g(y)k(a) = k(x+ya) \implies h = k$.

$\forall r$ in the roots of $g(p)[X]$, there exists some function which sends $a \rightarrow r$. Thus the canonical map is surjective.

Thus the map is bijection. $\square$

Problem 2b. Conclude $|\text{Ext}|$ is at most the degree of L/K , with equality iff $g(p)$ has all its roots in F and there are no double roots.

Proof. Thus $|\text{Ext}_K^L|$ is at equal the number of unique roots of $g(p)$ in F which is at most the number of unique roots of p which is at most $\deg p = [L : K]$. Thus $|\text{Ext}_K^L| \leq [L : K]$.

Suppose $|\text{Ext}_K^L| = [L : K]$. Then the number of unique roots of $g(p)$ in F is $[L : K]$. $\deg g(p) = \deg p = [L : K]$. Thus the number of unique roots of $g(p)$ in F is equal to the number of possible roots of $g(p)$ (or the degree). Thus every root is unique and $g(p)$ has no double root. Thus since the number of roots of $g(p)$ in F is equal to the number of roots of $g(p)$, all roots are in F . Thus $g(p)$ has all roots in F and no double roots.

Suppose $g(p)$ has all roots in F and no double roots. The number of unique roots of $g(p)$ in F is equal to $\deg g(p) = \deg p = [K : L]$. Since $|\text{Ext}_K^L|$ equals the number of number of unique roots of $g(p)$ in F , $|\text{Ext}_K^L| = [K : L]$. $\square$

Problem 3. Suppose we have $K \subseteq L \subseteq M$, all fields. Note there is a natural map $\alpha : \text{Ext}_K^M(g) \rightarrow \text{Ext}_K^L(g)$. Show that the fibre over q is exactly $\text{Ext}_L^M(q)$.

Proof. $\alpha^*(q) := \{h \in \text{Ext}_K^M(g) \mid \forall l \in L, h(l) = q(l)\} = \{h \in \text{Ext}_K^M(g) \mid h|_L = q\} = \text{Ext}_L^M(q)$, by the definition of ext . $\square$

Problem 4. Now show $\text{Ext}_K^L(g)$ has order at most degree L/K , with equality iff for every a in K , $g(p_a)$ has all its roots in F and these are all distinct, where p_a means the minimal monic for a in $K[X]$.

Proof. Let $L_i := L_{i-1}[\alpha]$ for some $\alpha \in L_i$. Let $L_0 = K$ and let there be some $N \in \mathbb{N} \ni \forall m \geq N, L_m = L$.

Base Case: $L_1 = K[\alpha]$. By problem 2b, $\text{Ext}_{L_0}^{L_1}(g) \leq [K[\alpha] : K]$ with the needed equality condition.

Inductive Hypothesis: Suppose $\text{Ext}_K^{L_{n-1}}(g)$ satisfies the theorem.

Inductive Step: $L_n = L_{n-1}[\alpha]$ for some $\alpha \in L_n$. Take $r : \text{Ext}_K^{L_n}(g) \rightarrow \text{Ext}_K^{L_{n-1}}(g)$. By the inductive hypothesis, $\text{Ext}_K^{L_{n-1}}(g) \leq [L_{n-1} : K]$. Furthermore from 3, we know $\forall q \in \text{Ext}_K^{L_{n-1}}(g)$, $r^*(q) = \text{Ext}_{L_{n-1}}^{L_n}(q)$

Let p be the minimal polynomial of α over L_{n-1} .

By 2b, $\text{Ext}_{L_{n-1}}^{L_n}(q) \leq [L_{n-1} : L_n]$ with equality when $q(p)$ has all its roots in L_i with no double roots.

Thus, $|\text{Ext}_K^{L_n}(g)| = \sum_{q \in \text{Ext}_K^{L_{n-1}}(g)} |r^*(q)| \leq |\text{Ext}_K^{L_{n-1}}(g)| |\text{Ext}_{L_{n-1}}^{L_n}(q)| \leq [L_{n-1} : K][L_n : L_{n-1}] = [L_n : K]$. Thus $|\text{Ext}_K^{L_n}(g)| \leq [L_n : K]$

Suppose, $\forall a \in L_n$, $p_a \in K[X]$, the minimum polynomial for a , $g(p_a)$ has all its roots in F and they are all distinct. Since $K \subseteq L_{n-1} \subseteq L_n$, then the minimal polynomial of a in L_{n-1} , p'_a , divides p_a . Since $g(p_a)$ has all its roots in F and the roots are distinct, $q(p'_a)$ must also have all its roots and they must be distinct. Thus $\text{Ext}_{L_{n-1}}^{L_n}(q) = [L_n : L_{n-1}]$. Since $L_{n-1} \subseteq L_n$, by induction hypothesis, $|\text{Ext}_K^{L_{n-1}}(g)| = [L_{n-1} : K]$. Thus, r is surjective since every element in $\text{Ext}_K^{L_{n-1}}(g)$ has the same fibre. Thus, since $\text{Ext}_K^{L_n}(g) = \sum_{q \in \text{Ext}_K^{L_{n-1}}(g)} |r^*(q)| = \sum_{q \in \text{Ext}_K^{L_{n-1}}(g)} [L_n : L_{n-1}] = [L_{n-1} : K][L_n : L_{n-1}] = [L : K]$

Suppose, $\text{Ext}_K^{L_n}(g) = [L : K]$. $\forall a \in L_n$. $K \subseteq K[a] \subseteq L_n$. Thus, $\text{Ext}_K^{K[a]}(g) = [K[a] : K]$. Thus $g(p_a)$ factors completely over F with distinct roots.

Thus the theorem is true for $\text{Ext}_K^L(g)$ by induction. $\square$

Problem 5. Now go back to the original problem, with $G := \text{Gal}(L/K)$. Show $|G|$ is at most the degree with equality iff for every a in L , its monic min polynomial over $K[X]$ has all its roots in L , and these are all distinct.

Proof. Suppose $F = L$ and $g = \text{id}$. By 1, $G = \text{Ext}_K^L(\text{id}_K)$. By problem 4, $|G| = |\text{Ext}_K^L(\text{id}_K)| \leq [L : K]$ with equality iff for every a in L , its monic min polynomial over $K[X]$ has all its roots in L , and these are all distinct. $\square$

Problem 6. Show in char 0 that any irreducible polynomial has distinct roots (in any field extension).

Proof. Let p be the given irreducible with a root α . Assume the contrary, p has a repeated root r . Thus p' , derivative of p wrt x , also has a root at x , because the condition for a repeated root is $p(x) = 0$ and $p'(x) = 0$. Thus $\gcd(p, p') \neq 1 \in F$. However since f is irreducible, its divisors are f and units. Thus $\gcd(p, p') = p$. However in $\text{char } F \neq 0$, f' is zero (if f is constant which contradicts our assumption of irreducibility, or must have degree less than f . But then f cannot divide f' . Thus a contradiction occurs. Thus f has no repeated roots. $\square$